

10/13 grp mtg TTT

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Lagrangian fibration (& Fourier-Mukai equivalences)

My (ultimate) motivation

To extend the results of Bridgeland ('99)

FM trans for K3 & ellip
fib

$\times$ $\swarrow$ $\searrow$ sheaf module
of fiber $P^h(X) \rightarrow P^h(X)$

Bridgeland-Macriocia ('02)

FM trans
for ellip surf

on FM trans for elliptic surfaces and K3 fibrations
to the setting of Lagrangian fibrations of hyperKähler
manifolds

In particular, the Frame work I have in mind is J. Sawon ('05)

Twisted FM transforms, (which provides a natural way
to treat fibrations without sections by incorporating
gerbes and twisted sheaves.)

$\rightarrow$ next time

To prepare for that I'll first review the case of elliptic K3
best known examples of Lagrangian fibrations.

I'll recall several characteristic features of ellip K3
and then discuss how these ideas generalize to
irreducible holomorphic symplectic (LHS) in higher dim.
ref (review on HK geom & Lag fib)

1. D. Huybrechts $\triangleright$ cpt HK mtds in "CY mtd and
related geometries" ch 3.
2. $\triangleright$ Lectures on K3 surfaces
3. A. Beauville $\triangleright$ Holomorphic sym geom: a partial list.
4. D. Huy & E. Muri $\triangleright$ Lagrangian fib.

elliptic: $c K^3 \setminus \mathbb{C}$

Def $X: K^3$ surf

$\pi: X \rightarrow \mathbb{P}^1$ surj morph.
 s.t. geometric generic fiber is
 smooth (integral) curves of genus 1

L : fund line bdl $K_X \sim \pi^*(K_{\mathbb{P}^1} \otimes L)$
 $L = (\pi^* \pi_{*} \omega_X)^{-1}$
 Automatically paper $K_B \otimes L \cong \mathcal{O}_B$
 $\deg L = \chi(\mathcal{O}_X)$

$\Rightarrow (2g-2) + 2 = 0$
 $\Rightarrow g(B) = 0$

possibly non-reduced

Note • Singular fibers are unions of rational curves with the possible types classified by Kodaira.

(I_0 : sm, I_1 : nodal, II : cusp, III , IV , I_n , I_n^* , II^* , III^* , IV^*)

w/ $\sum e(F) = e(X) = 24$ (general $I_1 \times 24$)

sing Euler number. ($\exists$ sing fib?)

top.

condition of lattice

K^3 is ellip $\iff \exists$ non triv line bundle L w/ $L^2 = 0$

movable part: $\Rightarrow$ elliptic (isotropic)

$\rho(X) \geq 5 \Rightarrow \exists$ ellip fib. ($2 \leq \rho(X) < 5$)
 NS lattice big enough. may, or may not.

$\exists$ latticept in direction of $D^2 = 0$.

• ellip $K^3 \subset$ Moduli of K^3 w/ $dg = 2d$.

dense

hyper

codim 1
 hypersurf
 (Heegner divisor)

dim 19

$\rightarrow$ dim 19
 polarised

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ hyperbolic lattice.

$\exists$ section $\iff U \xrightarrow{\text{prim}} NS(X)$
 Jacobian type $\langle f, s \rangle$

e.g. Jacobian fib. $Jac(X/\mathbb{P}^1)$ (cpt'd torsion free)
 minimal model

trivial $\mathcal{O}_X$

grp str. $\Rightarrow$ Néron model

semi-stable grsch $Pic^0(C) \cong G_m$ $Pic^0(C_{\text{gen}}) \cong G_m$

• $\exists$ section $\Rightarrow$ reconstructed by Weierstrass eq. $Pic^0(\text{red}) \cong (G_m)^r$
 $\exists$ Mordell-Weil gr. $\rightarrow$ Shioda - Tate orig. jin extend

• A section $\Rightarrow$ J-torsor & classified Tate-Safarevic $Pr(J)/Pr(P)$

IHS & Lag fib ^{depends on.} viewpoint $hul(X) \cong Sp(n)$ (3)

Def IHS (or cpt HK)

X admits a closed non-deg two-forms σ of type (2,0) (holomorphic symplectic form)

If X is simply conn. and $H^{2,0}(X) \cong \mathbb{C} \langle \sigma \rangle$ irred.

Mod even dim $\rightarrow$ Bogomolov-Birville decomp upto fin. cover ^{strict CY. torus.}

$\dim=2 \Rightarrow K3$ single family up to det.

$\dim \geq 4 \Rightarrow$ at this time we know two families in each dim

$\cdot K3^{(n)}$

$\cdot$ gen. Kum

$\cdot$ two isolated families

$\cdot OG6$

$\cdot OG10$

few examples?

Def Lag fib

$X: \text{IHS} \cdot \dim 2n$

$\pi: X \rightarrow B$ proper surj Birational var.

s.t. gen fib F is Lagrangian i.e. F : conn of dim n
 $\sigma|_F = 0$

property $\circ$ sm fibers of π are cpx tori (Arnold-Liouville)

$\circ$ Lag submtl of cpt HK is prj (Voisin)

$\leadsto$ fibers are abelian var.

(pol = const B. Wieneck '15)

abelian fib rel dim = $\frac{1}{2} \dim X$.

Matsushita

$X: \text{HK}$, of dim $2n$ B (Kähler) $0 < \dim B < 2n$

whose gen. fibre is conn

$f: X \rightarrow B$ proper surj

$\Rightarrow f: X \rightarrow B$: Lag fib

B is Fano with same Hodge number as $\mathbb{P}^n$

more precisely $\beta: k[t, Q]$ -factorial, Fano

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$$\rho(\beta) = 1$$

$$\beta: sm \rightarrow \beta \simeq \mathbb{P}^n \text{ (Hwang)}$$

known case!

$n=2 \Rightarrow \mathbb{P}^2$
(Ou, Huy. Xu)

Conj (HK SYZ)
 L : nontriv. $\delta(L) : BBI =$ Beauville Bogomolov-Fujita;
gen. intersection form on KS

not big

$$\delta(L) = 0 \text{ \& } L: nef$$

$$\Rightarrow \exists f: X \rightarrow \mathbb{P}^k \text{ s.t. } L = f^* \mathcal{O}_{\mathbb{P}^k}(k) \text{ for some } k \geq 1$$

$KS^{sm} : OK$ (Bayer & Macri)
OK for known case

Twisted FM Transf. for
hol. sym funtlds

Section (Sawon)

hyperelliptic

eg $S \rightarrow (\mathbb{P}^2)^V$ double cover of the plane ramified
over a sextic δ

- $S: KS$ — pull back of generic line
- C : genus 2 in S $|C| \cong \mathbb{P}^2$
- $C \rightarrow \mathbb{P}^2$ $Z^d = M_H^S(0, [C], d-1)$
 - $\deg L = d$
 - $\chi(L) = d+1-g$
 - gen sm $S=2$
 - sy red. irrred.

$$Z^d \sim_{det} \text{Hilb}^2(S) \quad \text{Pic}^d(C/|C|)$$

but Z^0 : has section Z' : torsion of Z^0 sing fib
easy to treat

SYZ conj for CY

$$I\text{-}q \times L_{qj} \Leftrightarrow J/k\text{-}special\ L_{qj}$$

$$Q_j = w_j + i w_k$$

HK notation